Table 1: The case study uses these parameters and yields these results.

Quantity	Value	
VEHICLE		
Model	CF2X (Crazyflie 2.x, X layout)	
Mass, arm length	27 g, 39.7 mm	
Thrust coefficient k_f	3.16×10^{-10} N/rpm^2	
Torque coefficient k_m	7.94×10^{-12} N m/rpm^2	
Maximum rotor speed, thrust/weight	21 710 rpm, 2.25	
SIMULATION		
Simulator	gym-pybullet-drones 2.2.0	
Physics, control rate	1200 Hz, 400 Hz	
Gravity	9.80665 m/s^2	
Aerodynamic effects, damping	off	
Initial state	at rest on the ground, yaw 0	
CONTROLLER	tuned [default]; others at defaults	
`MOT_SPIN_MIN`, `_MAX`, `_THST_EXPO`	0, 1, 0 (linear)	
`MOT_THST_HOVER`, `_HOVER_LEARN`	0.4444, 0	
`ATC_ANGLE_MAX`, `ATC_INPUT_TC`	30°, 0.05 [0.1] s	
`PSC_NE_POS_P`, `_D_POS_P`, `_D_VEL_P`	1.5 [1], 1, 5 s^{-1}	
`PSC_NE_VEL_P`, `_I`, `_D`	6 [2], 1, 0.5 [0.25]	
`PSC_D_ACC_P`, `_I`, `_D`	0.05, 0.1, 0	
`ATC_ANG_RLL_P`, `_PIT_P`, `_YAW_P`	9 [4.5], 9 [4.5], 4.5 s^{-1}	
`ATC_RAT_RLL_P`, `_I`, `_D`, same `PIT`	0.135, 0.135, 0.0036	
`ATC_RAT_YAW_P`, `_I`, `_D`	0.3, 0.02, 0	
`ATC_ACC_R_MAX`, `_P_MAX`, `_Y_MAX`	1800 [1100], 1800 [1100], 270 °/s^2	
MISSION		
Course, half-diagonal	figure eight, 1.25 m	
Altitude at w_0 to w_7	0.8, 1.0, 1.2, 1.4, 1.6, 1.4, 1.1, 0.8 m	
`WP_SPD`, `WP_ACC`	1 m/s, 1.5 m/s^2	
`WP_RADIUS_M`, `WP_ACC_CNR`	0.1 [2] m, 0	
`WP_JERK`	5 [1] m/s^3	
Path profile	S-curve legs, blended corners	
Heading (`WP_YAW_BEHAVIOR`)	2, along the desired velocity	
Hold at w_7	3 s within 0.1 m	
BUILD		
C++ vehicle	Apple clang 17.0.0, `-O2`	
C++ options	`-ffp-contract=off -fno-builtin -fno-fast-math -ftrapping-math -cl-single-precision-constant`	
Lean vehicle, C backend	v4.33.0-rc1, `-O3`	
Lean options	`-ffp-contract=off -fno-builtin`	
Binary size, C++ and Lean	0.23 MB, 3.64 MB	
Time per tick, C++ and Lean	1.04 µs, 10.31 µs	
RESULTS	defaults	tuned
End of the run	time limit	hold at w_7
Flight time (ticks)	60.00 s (24000)	19.49 s (7797)
Final position error	17.58 cm	2.97 cm
Maximum |roll|, |pitch|	8.8°, 6.0°	14.8°, 11.5°
Maximum tilt	8.9°	16.2°
Tracking error (max, rms)	10.8, 4.1 cm	8.7, 2.3 cm
Closest approach to w_1 to w_6	10.0–15.7 cm	7.0–7.3 cm
Within 0.1 m of w_7	19.14 s	16.49 s
Round trip, C++ (median, p99)	20, 26 µs	20, 27 µs
Round trip, Lean (median, p99)	166, 190 µs	163, 203 µs
Identical ticks, Lean and C++	24000 of 24000	7797 of 7797